

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

Annexure A

Industry Problem Solving Task

Code of Conduct:

*I Harish Kumar / 192521340 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A

Industry Problem Solving Task

1. 1) Star Topology for a 20-Person Startup

Requirement Analysis

A 20-person startup on a single office floor needs a network that is inexpensive to set up, simple enough for a small (or non-existent) IT team to manage and troubleshoot, and flexible enough to accommodate future hires without a redesign. It must support standard office traffic — web browsing, file sharing, cloud application access — as well as VoIP phones for calls and Wi-Fi for laptops and mobile devices. Given the small scale, elaborate redundancy (as required for a bank in Question 2) would be unnecessary cost; the priority instead is ease of management and low cost per connection.

Why Star Topology

A star topology, where every device connects individually to a central switch, is the natural fit. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with headroom to grow. Each device connects to the switch using Cat6 UTP cabling, which supports Gigabit speeds up to 100 metres — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or a pfSense appliance) that handles internet access, DHCP, and network address translation, while a Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices.

Star Topology — 20-Person Startup Office

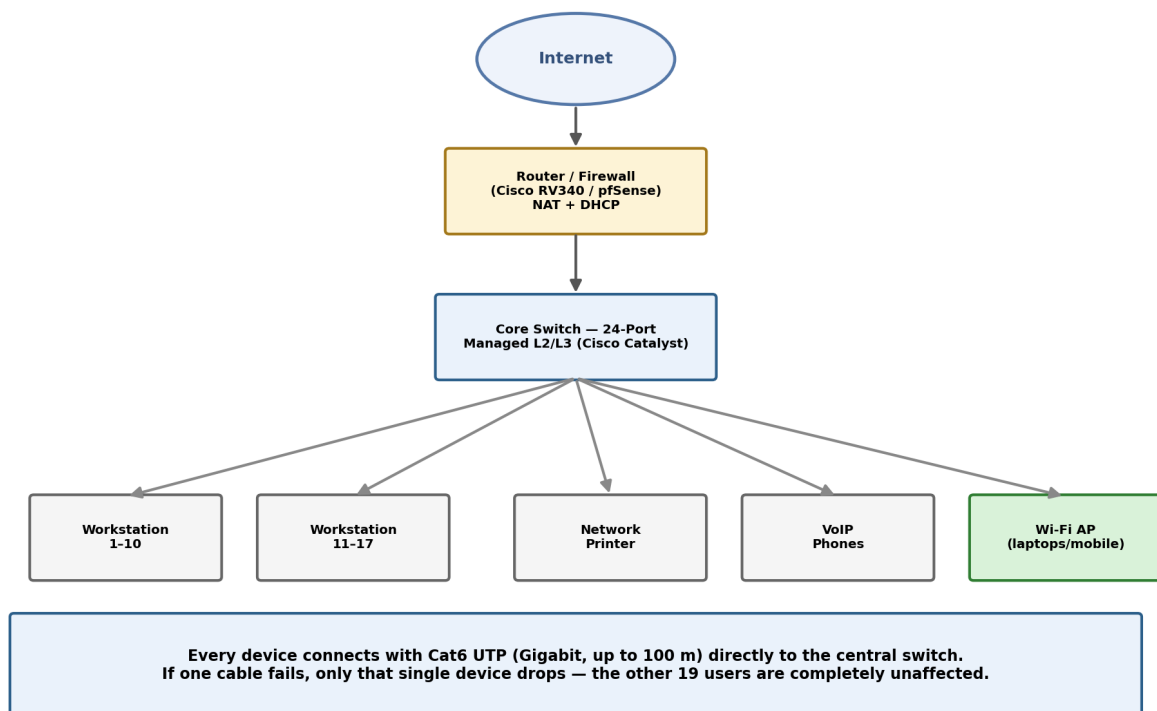


Figure 1.1 – Star topology network for a 20-person startup office.

Device & Cabling Specification

Component	Recommended Model/Type	Purpose	Approx. Qty
Core switch	24-port managed L2/L3 (Cisco Catalyst 1200 / Netgear GS724T)	Central connection point for all devices	1
Router/Firewall	Cisco RV340 or pfSense appliance	Internet access, DHCP, NAT, basic security	1
Wi-Fi Access Point	Cisco/Ubiquiti PoE AP	Wireless coverage for laptops/mobile	1–2
Cat6 UTP cabling	Solid-core, up to 100 m runs	Gigabit connection from switch to every device	~25 runs
VoIP handsets	SIP-compatible desk phones	Office telephony	As needed

Fault Isolation — the Key Advantage

The central advantage of this design is fault isolation: because each device has its own dedicated cable back to the switch, if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees completely unaffected. This is a direct structural improvement over a shared-medium design (see Question 3), where one bad cable can disrupt everyone at once. Centralized cable management — running all cabling to one wiring closet or rack — also greatly simplifies future moves, adds, and changes: relocating a desk is simply a matter of running (or re-patching) one new cable, without touching anyone else's connection.

Fault Isolation in a Star Topology

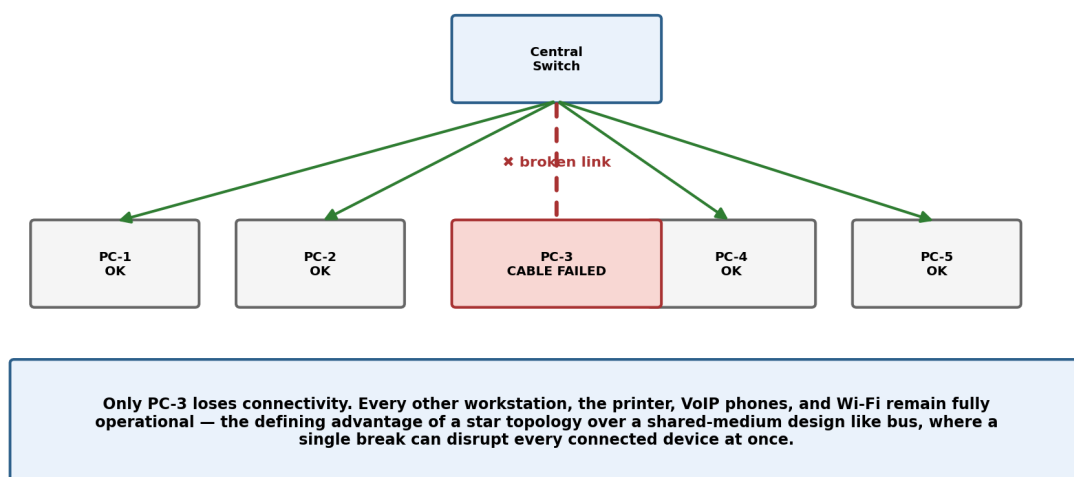


Figure 1.2 – A single cable failure only isolates the one affected device.

Scalability & Cost Notes

Choosing a 24-port switch (rather than exactly matching today's headcount) leaves 4–8 free ports for near-term growth without new hardware. Because the core switch is Layer 2/3 capable, the network is also VLAN-ready from day one — if the startup later wants to separate, say, engineering traffic from guest Wi-Fi (as elaborated in Question 5's departmental design), it can

do so with a configuration change rather than a rewiring project. Overall hardware cost for this scale (one managed switch, one router/firewall, one or two APs, and Cat6 cabling for ~20 drops) is modest compared to the mesh or hierarchical designs required at enterprise scale, making star topology the most cost-effective correct choice for this size of office.

2. **2) Mesh Topology for a Bank**

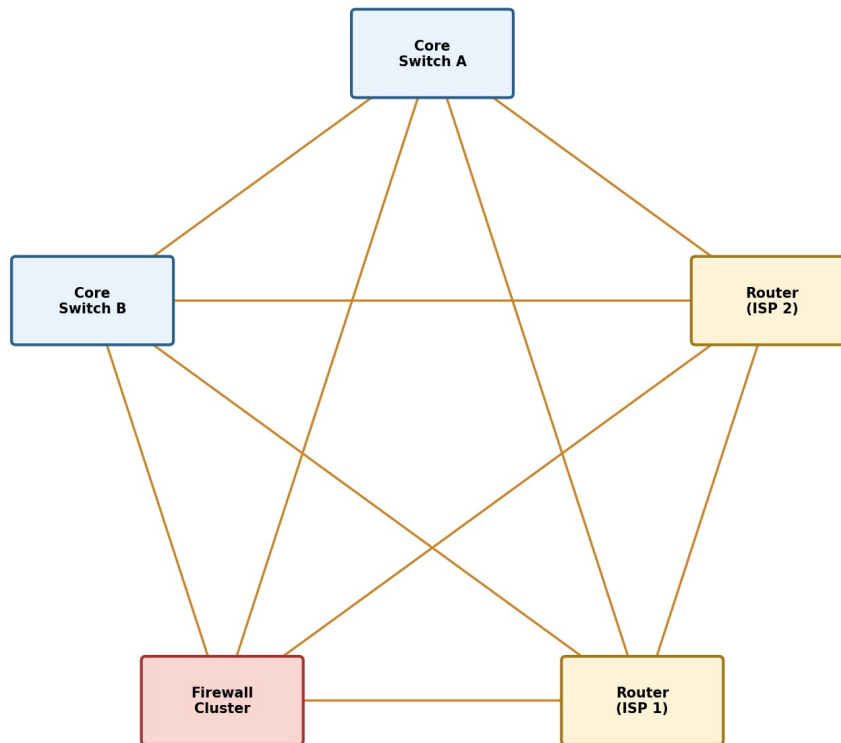
Requirement Analysis

A bank's network must satisfy near-zero downtime, strict regulatory requirements for uptime and data integrity, and the low-latency demands of financial transaction processing, where even brief outages can mean failed trades, compliance violations, and direct financial loss. This rules out any design with a single point of failure, which immediately points toward a mesh-based core.

Why (Partial) Mesh

In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For n core devices, a full mesh requires $n(n-1)/2$ direct links; for example, 5 core devices need 10 links, as shown below. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other), while the distribution and access layers use redundant uplinks to at least two core switches — giving nearly all of full mesh's resilience at a fraction of the cabling cost, since only the highest-value core layer needs the full $n(n-1)/2$ treatment.

Full Mesh — Bank Core Layer (Every Device ↔ Every Device)



5 devices → $n(n-1)/2 = 10$ direct links: every core device reaches every other core device in exactly one hop — no single link or node failure isolates any part of the core.

Figure 2.1 – Full mesh at the bank's core layer — every device reaches every other in one hop.

Device & Cabling Specification

- **Dual enterprise-grade routers:** e.g. Cisco ASR or Juniper MX series, providing BGP-based redundant connectivity to two separate ISPs.
- **Core switches:** enterprise-class (e.g. Cisco Nexus/Catalyst 9500) interconnected in a full mesh via fiber.
- **Fiber links between core devices:** single-mode fiber for long distances between buildings, multi-mode OM4 within the data center — fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for low-latency financial transactions.
- Redundant power supplies and UPS systems on all core devices, plus RAID storage arrays for data resilience.
- **Distribution/access layers:** redundant uplinks from each switch to at least two core switches, rather than a single point of connection.

Bank Network — Redundant Core with Dual-Homed Access

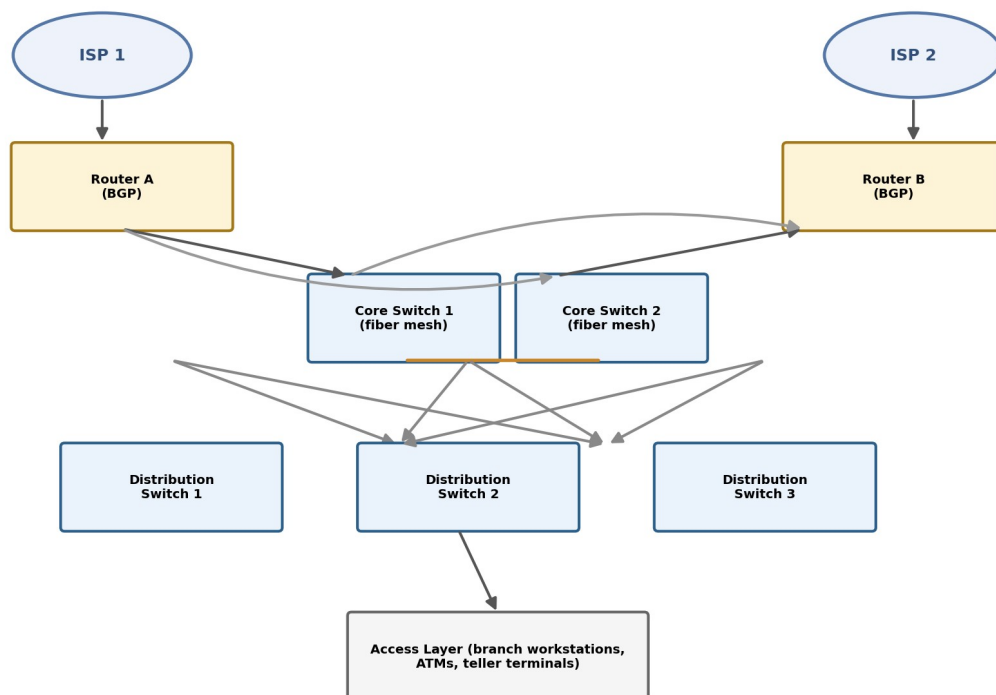


Figure 2.2 – Dual-ISP, dual-router, mesh-core design with redundant distribution/access uplinks.

Automatic Failover Justification

The justification for this design is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using dynamic routing protocols like OSPF (within the core) or BGP (across the dual ISP connections). Neither administrators nor applications need to intervene — the routing protocols detect the failed link (via missed hellos/keepalives) and recompute a path through a surviving link almost immediately, satisfying regulatory requirements for uptime (banks are frequently held to standards approaching 99.999% availability) and for data integrity, since financial transactions cannot be allowed to silently fail mid-transit.

Cost vs. Downtime Trade-off

Mesh topology is unquestionably the most expensive design to build and cable, due to the sheer number of links and the premium cost of fiber, enterprise-grade hardware, and redundant power. For a bank, however, this cost is justified: the cost of even a short outage — in lost transactions, regulatory penalties, and reputational damage — far exceeds the incremental infrastructure cost of full/partial mesh redundancy, which is precisely why this level of investment is standard practice in the financial sector despite being overkill for a business like the startup in Question 1.

3. 3) Bus Topology for a Classroom Lab

Requirement Analysis

A classroom lab has a small, fixed set of computers, a tight budget, and light, supervised usage — mainly internet browsing and file sharing rather than mission-critical traffic. There is no need for

the redundancy of a mesh or even the centralized management overhead of a full star deployment with a dedicated managed switch; the priority here is minimizing hardware cost.

Why Bus Topology

Bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub entirely. A single coaxial cable (RG-58, 50-ohm) runs along the length of the room; in the classic implementation each computer connects to this backbone using a T-connector and a BNC connector, with 50-ohm terminators placed at both ends of the cable to absorb signals and prevent reflection.

Classic Bus Topology — Coaxial Backbone (RG-58)

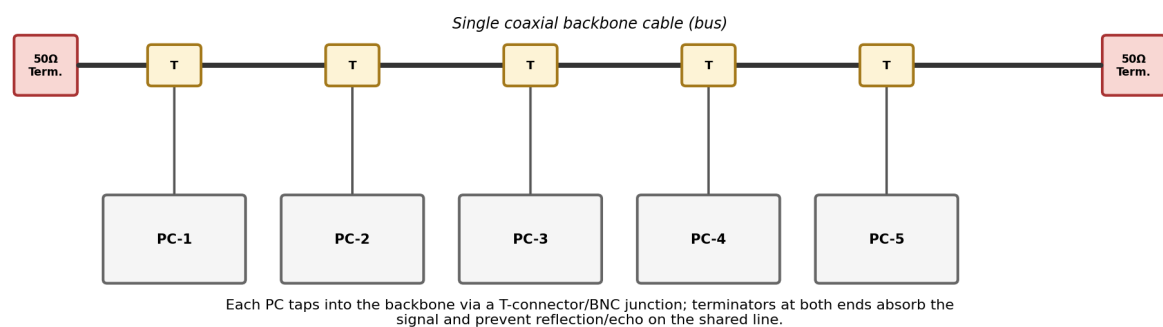


Figure 3.1 – Classic coaxial bus topology with T-connectors and terminators.

Modern Low-Cost Variant

For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the coaxial backbone and terminators, using Cat5e UTP patch cables to each workstation — achieving bus-like simplicity and cost at minimal expense, using hardware that is far easier to source and maintain today than legacy coaxial components.

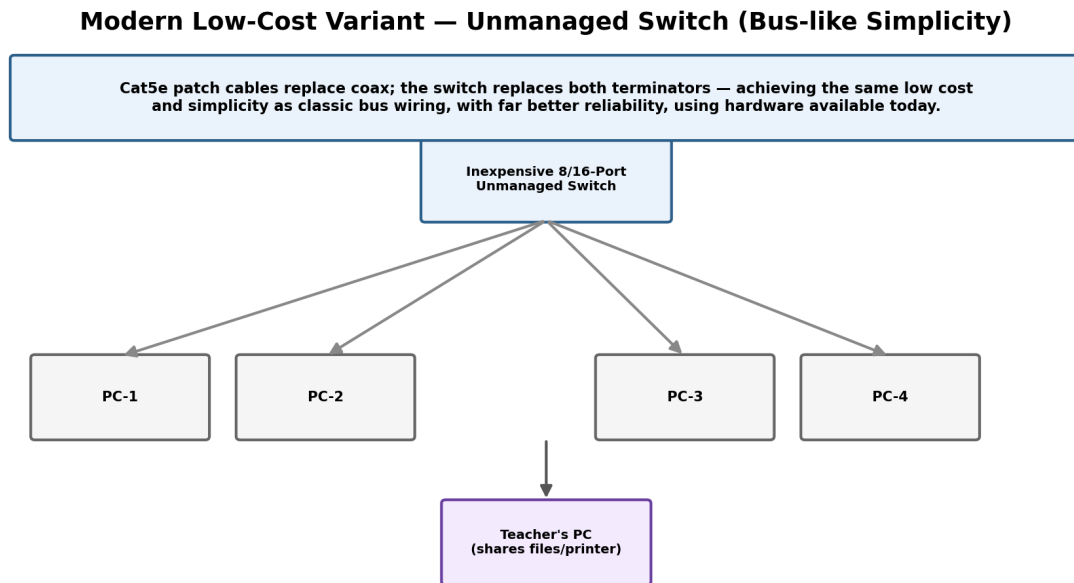


Figure 3.2 – Modern equivalent using a low-cost unmanaged switch.

Limitations & Trade-offs

The primary limitation of a true bus design is that the shared medium is susceptible to collisions (multiple computers transmitting simultaneously on the same line), and a single cable break disrupts all connected machines at once — the opposite of the fault isolation seen in the star design of Question 1. For a supervised classroom environment with light, non-critical usage, however, this trade-off is entirely acceptable: the cost savings and simplicity outweigh the risk of an occasional, easily-diagnosed, single-cable outage. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file and printer sharing.

Aspect	Classic Coax Bus	Modern Unmanaged-Switch Variant
Hardware cost	Lowest (coax + connectors + terminators)	Very low (one cheap switch + Cat5e cables)
Maintainability	Harder — legacy components, awkward to extend	Easier — standard, widely available parts
Fault behaviour	One break disrupts entire bus	One cable failure isolates only that PC
Recommended today	Historical/legacy reference	Preferred in practice

4. 4) Hybrid Topology for a University Campus

Requirement Analysis

A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — each with different traffic patterns and criticality, making a single topology inadequate on its own. The network must be redundant at the backbone (a core outage cannot be allowed to isolate an entire building), manageable at the building level, cost-efficient within individual labs, and above all scalable, since campuses regularly add new buildings and expand existing ones.

Why a Hybrid (Hierarchical) Topology

A hybrid topology combines the best elements of multiple designs: a fiber-based ring or partial mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and simple star/tree structures within individual labs for cost-efficiency (the same reasoning as Question 3, applied inside each lab room). At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches in a ring or partial-mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks; within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cabling.

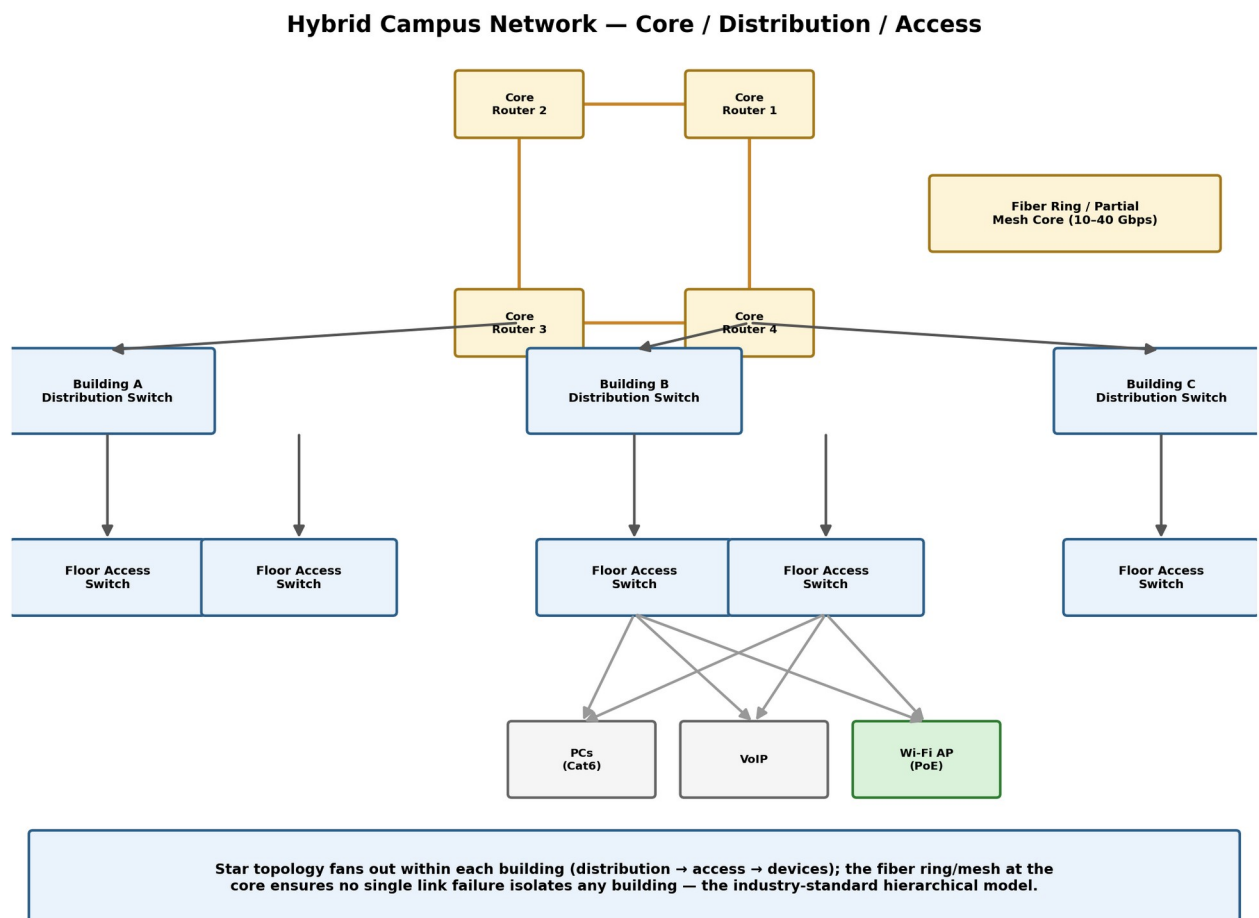
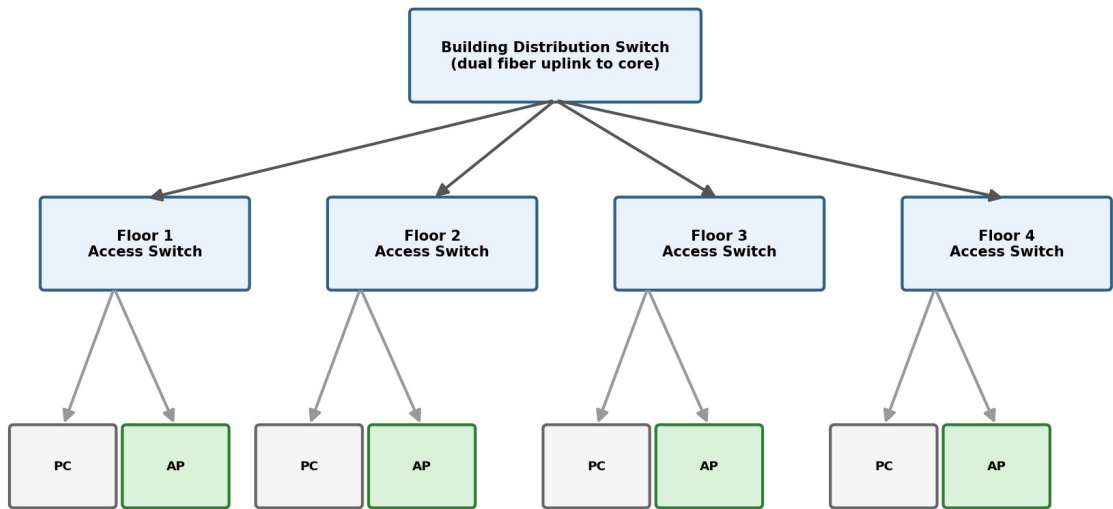


Figure 4.1 – Full hybrid campus network: fiber ring/mesh core, per-building distribution, per-floor access.

The Three-Layer Hierarchical Model

This hierarchical three-layer model — core, distribution, and access — is the industry standard for enterprise and campus networks precisely because it isolates change: adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to any existing building's infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling to every AP location.

Zoomed View — Star Fan-Out Within a Single Building



Wireless access points connect via PoE — no separate power cabling needed.

Figure 4.2 – Zoomed view: star fan-out from a building's distribution switch down to end devices.

Summary — Topology by Layer

Layer	Topology Used	Reason
Core (campus backbone)	Fiber ring / partial mesh	No single link failure isolates a building; very high bandwidth (10–40 Gbps)
Distribution (per building)	Dual fiber uplinks to core	Redundant path from each building to the backbone
Access (per floor/lab)	Star (and low-cost bus/tree in labs)	Simple, manageable, cost-efficient at the device level

The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide — exactly matching the layered requirements of a large, multi-building institution that none of the single topologies in Questions 1–3 could satisfy on their own.

5. 5) VLAN-Based Multi-Department Network

Requirement Analysis

A company with multiple departments — such as HR, Finance, Sales, and IT — needs an architecture that lets devices communicate efficiently within a department, allows controlled communication between departments, and keeps each department's traffic logically separated for both security and performance (e.g. HR and Finance handle sensitive data that should not be broadly visible to every other department by default).

Design — VLANs over a Layer 2/Layer 3 Hierarchy

Each department is connected through dedicated Layer 2 access switches, allowing devices like computers and printers to communicate within the same department. These access switches then connect to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic — broadcasts from the Sales VLAN, for instance, never reach HR or Finance devices at all, regardless of physical cabling.

VLAN-Segmented Multi-Department Network

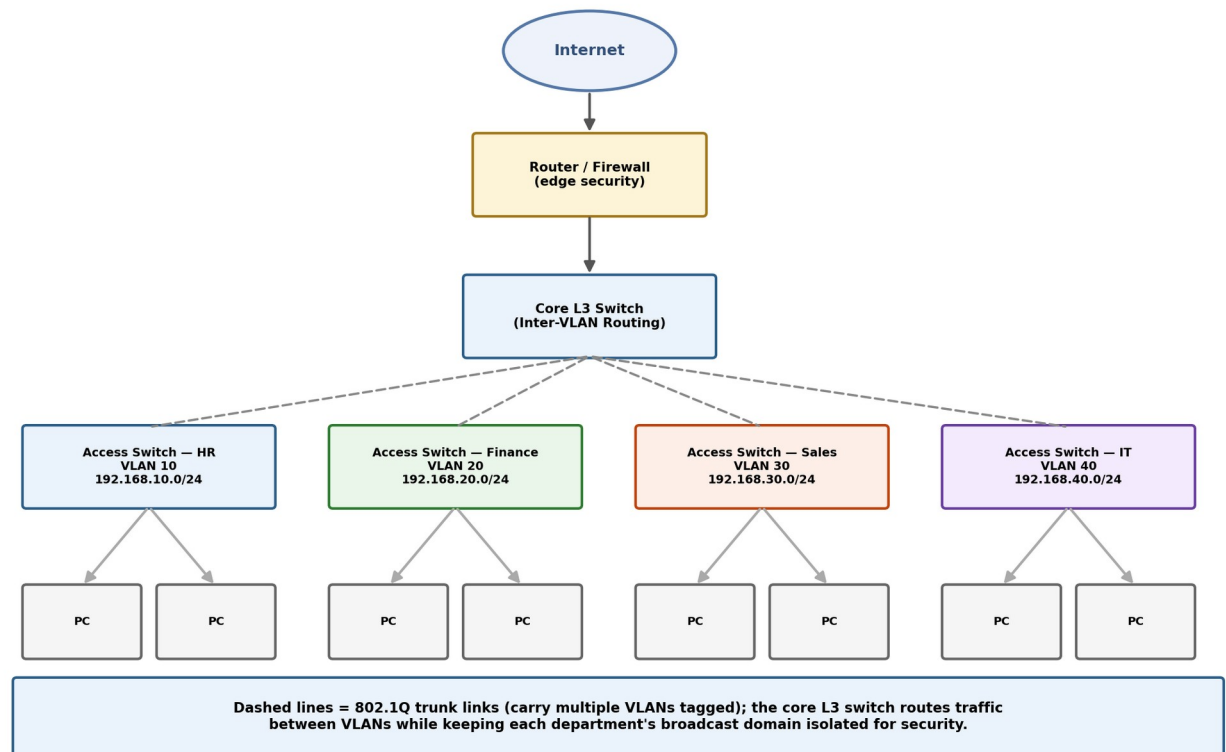


Figure 5.1 – Four departments on separate VLANs, trunked to a core Layer 3 switch for inter-VLAN routing.

IP Addressing Plan

A clean, department-aligned subnetting scheme makes the network easy to manage, secure with per-VLAN access control lists, and simple to extend:

Department	VLAN ID	Subnet	Access Switch
HR	10	192.168.10.0/24	Dedicated L2 access switch
Finance	20	192.168.20.0/24	Dedicated L2 access switch
Sales	30	192.168.30.0/24	Dedicated L2 access switch
IT	40	192.168.40.0/24	Dedicated L2 access switch

Internet Access, Security & Redundancy

A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security — all inter-VLAN traffic and internet-bound traffic passes through policy enforcement points rather than flowing unrestricted. Redundant links and backup paths can be included (e.g. dual uplinks from access switches to the core, with spanning-tree or link-aggregation protocols managing failover) to ensure reliability if a primary link fails.

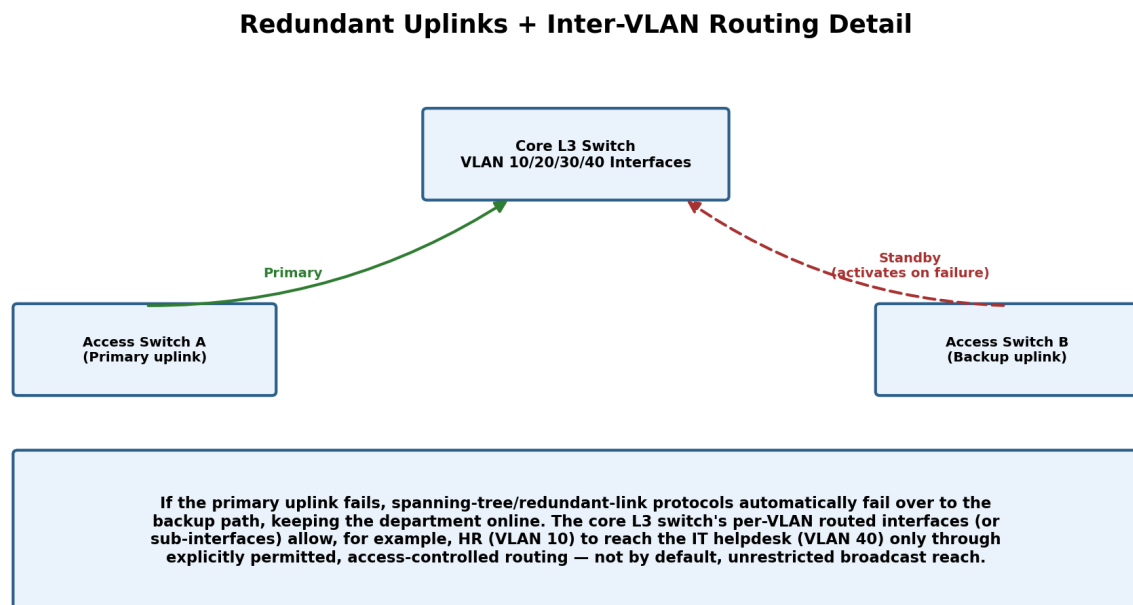


Figure 5.2 – Redundant access-switch uplinks and controlled inter-VLAN routing.

Why This Design Scales

This hierarchical, VLAN-segmented design improves performance (each department's broadcast domain stays small), enhances security (department traffic is isolated by default, with inter-VLAN access explicitly controlled at the core router/firewall), and allows easy expansion as the company grows — adding a new department is simply a matter of provisioning a new VLAN and subnet on existing switches (or adding one more access switch), without redesigning the network, in the same spirit of incremental scalability seen in the campus design of Question 4.